Devin Wong

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Education

University of California, San Diego **Expected June 2028**
Bachelor of Science, Mechanical Engineering *La Jolla, CA*
Available for an internship from **June 2027** to **September 2027**

Experience

Tesla (Incoming) **Sep 2026 – Jun 2027**
Tooling & Process Engineering Intern - Castings *Lathrop, CA*

- Responsible for Engineering of Cast and Trim dies for High Pressure Die-cast components
- Design for optimized runner & gate, thermally balanced inserts, and highest possible vacuum efficiency
- Evaluate molten aluminum profiles and gating geometry via flow simulations (Magmasoft & Flow 3D)
- Employ dimensional analysis, tolerance stack-ups, and GD&T to improve dimensional capabilities of castings

Roche **Jun 2026 – Sep 2026**
Mechanical Engineering Intern - Sensor Module Development *Santa Clara, CA*

- Evaluated the impact of fluidic geometry on shear stress distributions and flow impedance in sequencing hardware
- Conducted steady-state CFD analysis in **SolidWorks Flow Simulation** to model flow across microfluidic geometry
- Performed steady-state thermal analysis across sensor interfaces to improve temperature uniformity by a **factor of two**
- Designed and fabricated fixtures for pipette tip positioning and developed test stands for fluidic interface prototypes

Triton Racing (Formula SAE) - University of California, San Diego **Oct 2025 – Jun 2026**
Mechanical Design & Analysis Engineer - Aerodynamics *San Diego, CA*

- Designed the TR26 rear wing aerodynamic assembly to maximize downforce and optimize structural load paths
- Performed structural FEA in **ANSYS Mechanical** validating parts under **200+ lb** distributed loads (FOS ∼**1.5**)
- Conducted 2D steady-state CFD in **ANSYS Fluent** to optimize pressure distributions, slot gaps, and element sizing
- Generated CAM toolpaths in **Fusion 360** to machine high-density foam molds on a 3-axis CNC router
- Authored **10+** detailed **GD&T** drawings (ASME Y14.5-2018), improving fabrication and alignment accuracy

Projects

Roche | Sensor Module Architecture Optimization | *Fluid Mechanics, Mechanical Testing, Data Analysis* **Jun 2026**

- Conducted parametric trade studies on 4 microfluidic sensor module geometries for research and early development
- Identified the design that reduced wall shear stress non-uniformity by **70%** and pressure drop by **72%**
- Built steady-state CFD flow simulations utilizing grid-independence studies with up to **15 million** cell meshes
- Developed a precision testing setup using a PD pump and pressure sensor to physically validate CFD trends

Triton Racing | Rear Wing Spar | *SolidWorks, ANSYS ACP, Fusion 360, CNC Router, Wet Layup* **Jan 2026**

- Designed a **31"** carbon-fiber spar carrying a **400 N/m** distributed aerodynamic design load, validated to a FOS of **5.5**
- Created CAM toolpaths to CNC-machine a foam mold to **0.004**" stepdown tolerance, enabling clean release post layup
- Conducted a **4-ply** bidirectional carbon-fiber layup holding peak deflection to **0.38 mm**
- Validated laminate stiffness against ANSYS Composite Analysis with a physical **3-point bend test**

Triton Racing | Rear Wing Mounts | *SolidWorks, ANSYS Mechanical, FEA, Bolt Theory, Water Jet* **Nov 2025**

- Designed **6061-T6** aluminum swan-neck mounts using topology optimization, reducing intial weight by **77%**
- Hand-calculated bending stress and deflection under a **200N** load case for each mount as a baseline
- Designed a **2.2 million cell** FEA convergence study (<**0.5%** stress variation), verifying a **11.43 mm** displacement
- Conducted bolt shear and bearing strength analysis at the chassis interface, validating a static FOS of **47.53**

Technical Skills

Software: SolidWorks, SolidWorks Flow Simulation, ANSYS Mechanical, ANSYS ACP, MATLAB, Autodesk Fusion 360
Design: GD&T (ASME Y14.5-2018), DFM/DFA, CAD, CAM, FEA, CFD
Machining: CNC Router, Laser Cutter, 3D Printer, Water Jet, Hot-Wire Cutter